In-Depth Reading Analysis Report

Paper: *Training-Free LLM-Generated Text Detection by Mining Token Probability Sequences* (Xu et al., ICLR 2025)

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1. Core Motivation

1.1 Fundamental flaws of training-based detectors (Section 1, Section 2)

The paper identifies two structural weaknesses of training-based detectors (e.g., RoBERTa-based fine-tuning, OpenAI Text Classifier, GPTZero):

- **High resource cost.** They "require significant time and effort in high-quality data collection, model training, etc." (Section 1).
- **Weak generalization / overfitting.** They "often face generalization issues, e.g. overfitting to in-domain data, particularly when detecting cross-domain texts" (Section 1, citing Pu et al., 2023; Zhu et al., 2023). Related work restates this: many training-based methods "struggle with overfitting or generalizing to out-of-distribution data" (Section 2, citing Uchendu et al., 2020; Chakraborty et al., 2023).

1.2 Fundamental flaws of existing training-free detectors (Abstract, Sections 1–2)

The paper attributes the limitations of existing training-free detectors (Likelihood, Rank, GLTR, DetectLRR, DetectGPT, DetectNPR, DNA-GPT, Fast-DetectGPT) to two issues:

- **Neglect of local statistics.** They "typically rely on global text sequence statistics, neglecting the modeling of local discriminative features, thereby limiting their detection efficacy" (Abstract). Most methods "only evaluate the global statistical features (e.g. likelihood) of the text without harnessing the local statistical patterns (e.g. temporal dynamics) from token segments" (Section 1). The authors state their hypothesis explicitly at the end of Section 2: the performance limits of training-free detectors in black-box detection "arise from their failure to fully exploit or excavate the deeper layers of information inherent in token probability sequence, particularly the local geometric information."
- **Efficiency–efficacy dilemma.** Contrast-sample-based methods (DetectGPT, DetectNPR, DNA-GPT) "cannot achieve real-time or large-scale detection due to their excessive time consumption" (Section 2). Fast-DetectGPT improved efficiency (340× over DetectGPT), but "its black-box detection performance still exhibits substantial potential for enhancement" (Section 2). In short: existing training-free methods are *either* costly *or* weak under cross-model/black-box conditions and paraphrasing attacks (Section 1).

1.3 The authors' breakthrough: time-series analysis + local statistical features (Abstract, Section 1, Figure 1)

- The paper "for the first time ... introduce[s] time series analysis to LLM-generated text detection, capturing the temporal dynamics of token probability sequences" (Abstract). The TPS is treated as a **time series**, where token position plays the role of time.
- The empirical premise is Figure 1(a): under the same prefix, "the TPS of human-written text fluctuates more abruptly than LLM-generated ones, showcasing distinct temporal dynamics" (Section 1).
- To quantify this, the authors borrow **Diversity Entropy (DE)** (Wang et al., 2020), an entropy-based measure of dynamical complexity originally designed for fault diagnosis, ECG, and signal processing, and re-purpose it

to mine **local** fluctuation statistics from the TPS at multiple time scales. These local features are then fused with the **global** statistic (log-likelihood), producing a single discriminative score (Figure 1(b), Figure 2).

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2. Methodology Breakdown

2.1 Temporal characteristics of the Token Probability Sequence (TPS) (Section 3.1, Eq. 1)

For a candidate text $t = (t_1, t_2, \dots, t_n)$ and a scoring (proxy) model M_θ, the TPS is the positional sequence of conditional log-probabilities:

$$L(t) = \left(\log p_\theta(t_1 \mid t_{<1}),\, \log p_\theta(t_2 \mid t_{<2}),\, \dots,\, \log p_\theta(t_n \mid t_{<n})\right) \tag{1}$$

Its key temporal properties, as argued in the paper:

- **Human texts fluctuate abruptly; LLM texts are smoother/more compact.** The token probabilities of LLM-generated continuations "are more compact, whereas human responses exhibit the opposite trend," which the authors attribute to LLMs being "trained with the objective of minimizing perplexity" (Appendix E, Figure 1(a)).
- **Scale-dependent regularity.** After multi-scale smoothing, "there is an overall downward trend in the DE values ... as the scale increases" for *both* classes — repeated averaging drives the sequence "gradually approach a periodic or deterministic state" (Appendix E, Figure 9).
- **Cross-conditions consistency.** The human DE curve lies *above* the LLM curve at every scale, "and this conclusion holds across various languages, scenarios, and models" (Appendix E, Figure 9). This is the justification for using local-dynamics features for detection.

2.2 Diversity Entropy (DE) and Multi-scale Diversity Entropy (MDE) (Section 3.1, Eqs. 2–8, Figure 2)

DE is described as "a process of first increasing the dimensionality and then reducing the dimensionality of a one-dimensional sequence" (Section 3.1). The five-stage pipeline is:

1. **Multi-scale transformation (coarse-graining).** For scales $\tau = 1, 2, \dots, \tau'$, compute the moving-average TPS:

$$\log p_\theta^{(\tau)}(j) = \frac{1}{\tau}\sum_{i=j}^{j+\tau-1} \log p_\theta(t_i \mid t_{<i}), \quad 1 \le j \le n-\tau+1 \tag{2}$$

$\tau = 1$ recovers the original TPS. Different τ's capture dynamics at different scales — "essentially an enhancement of the original data."

2. **Sliding-segmentation.** For each scale, a window of size s with stride 1 segments $L^{(\tau)}(t)$ into rows of a matrix $\tilde{L}^{(\tau)}(s)$ (Eq. 3), whose i-th row is the *s-probability segment*

$$l_i^{(\tau)}(s) = \left(\log p_\theta^{(\tau)}(i),\, \log p_\theta^{(\tau)}(i+1),\, \dots,\, \log p_\theta^{(\tau)}(i+s-1)\right) \tag{4}$$

3. **Segment similarity.** The cosine similarity of *adjacent* segments yields the similarity sequence $D^{(\tau)}(s)$, with

$$d_k^{(\tau)} = \mathrm{CosineSimilarity}\left(l_k^{(\tau)}(s),\, l_{k+1}^{(\tau)}(s)\right), \quad k = 1, \dots, n-\tau-s+1 \tag{5}$$

Geometric interpretation (Section 3.1): the closer $d_k^{(\tau)}$ is to 1, the smaller the angular deviation between adjacent segments, i.e., the **smaller the local fluctuation**; conversely, values farther from 1 indicate larger local fluctuation.

4. **Probability state sequence.** The interval [-1, 1] is partitioned into ε equal, mutually exclusive sub-intervals ("states"). The histogram of $D^{(\tau)}(s)$ over these states, normalized, gives the probability state sequence

$$P^{(\tau)}(s) = \left(P_1^{(\tau)}, \dots, P_\varepsilon^{(\tau)}\right), \quad P_i^{(\tau)} = \frac{c_i^{(\tau)}}{\sum_{k=1}^{\varepsilon} c_k^{(\tau)}} \tag{6}$$

5. **Normalized Shannon entropy.** For each scale:

$$\mathrm{DE}(s, \varepsilon, \tau) = -\frac{1}{\ln \varepsilon}\sum_{i=1}^{\varepsilon} P_i^{(\tau)} \ln P_i^{(\tau)}, \quad P_i^{(\tau)} > 0 \tag{8}$$

Because of the $\frac{1}{\ln\varepsilon}$ normalization, $\mathrm{DE} \in [0, 1]$. A DE closer to 0 means the TPS at that scale has **less fluctuation** (more regular), while a value further from 0 means **greater fluctuation** (Section 3.1).

The **MDE sequence** stacks DE across scales:

$$\mathrm{MDE}(t, s, \varepsilon, \tau') = \left(\mathrm{DE}(s,\varepsilon,1), \dots, \mathrm{DE}(s, \varepsilon,\tau')\right) \tag{7}$$

This is precisely how the method captures **local** fluctuations: cosine similarity measures direction change between neighboring short segments (local geometry), the histogram-entropy converts the similarity distribution into a complexity measure, and the multi-scale layer aggregates dynamics from fine to coarse granularity.

2.3 The Lastde score: fusing global and local statistics (Section 3.2, Eq. 9)

The **Lastde** score (Log-Likelihood **and** sequential token probability diversity entropy) is a ratio:

$$\mathrm{Lastde}(t, \theta) = \frac{\dfrac{1}{n}\sum_{i=1}^{n}\log p_\theta(t_i \mid t_{<i})}{\mathrm{Agg}\left(\mathrm{DE}(s,\varepsilon,1), \dots, \mathrm{DE}(s,\varepsilon,\tau')\right)} \tag{9}$$

- **Numerator** = average Log-Likelihood — the *global* statistic of the TPS.
- **Denominator** = "Agg-MDE" — the *local* statistic. $\mathrm{Agg}: \mathbb{R}^{\tau'} \to \mathbb{R}$ defaults to the **standard deviation (Std)** over scales (alternatives 2-Norm, Range, ExpRange, ExpStd are analyzed in Appendix D, Table 8).
- Default hyperparameters: $s = 3$, $\varepsilon = 10 \times n$, $\tau' = 5$ (Section 3.2). (For Lastde++, $s=4$, $\varepsilon = 8\times n$, $\tau' = 15$.)
- **Decision rule:** score versus threshold. Human-written texts generally yield *lower* Lastde scores and LLM-generated texts *higher* ones (Section 3.2, Figure 1(b), Figure 11); the approximate separating threshold reported in Appendix E is -40.

- The detection process has three steps (Section 3.2, Figure 2): **(1) Inference** — get the TPS from an open-source proxy model; **(2) Computing statistics** — Log-Likelihood and Agg-MDE computed in parallel; **(3) Scoring** — divide the two and threshold.
- The ablation (Appendix E, Table 11, Figure 10) confirms the fusion is essential: Log-Likelihood or Agg-MDE alone give only weak separation, while their combination "can sometimes be improved by over 10%" in AUROC — the global measure summarizes whole-sequence information while Agg-MDE captures segment-level information, and the two "complement each other."

2.4 Lastde++: fast-sampling and sampling discrepancy (Section 3.2, Eqs. 10–11)

Lastde++ upgrades the single Lastde score with a **sampling-discrepancy** perspective, borrowing the fast-sampling technique from Fast-DetectGPT (Bao et al., 2024):

$$D\left(t, \theta, \tilde{\theta}\right) = \frac{\mathrm{Lastde}(t, \theta) - \tilde{\mu}}{\tilde{\sigma}} \tag{10}$$

with

$$\tilde{\mu} = \mathbb{E}_{\tilde{t} \sim M_{\tilde{\theta}}(\tilde{t} \mid t)}\left[\mathrm{Lastde}(\tilde{t}, \theta)\right], \qquad \tilde{\sigma}^2 = \mathbb{E}_{\tilde{t} \sim M_{\tilde{\theta}}(\tilde{t} \mid t)}\left[\left(\mathrm{Lastde}(\tilde{t}, \theta) - \tilde{\mu}\right)^2\right] \tag{11}$$

Key properties:

- $M_{\tilde{\theta}}$ is a **conditionally independent sampling model**: each new sample $\tilde{t}$ is drawn conditioned on the original text t. The mechanics of fast-sampling itself are not re-derived in the paper — it is adopted from Fast-DetectGPT (Bao et al., 2024), as stated in Section 3.2.
- The score is a **z-score**: the raw Lastde score is standardized by the mean $\tilde{\mu}$ and standard deviation $\tilde{\sigma}$ of Lastde scores of the sampled continuations.
- **Efficiency.** The default number of samples is **100**, which is 1/100 of Fast-DetectGPT's 10,000 (Section 3.2, Appendix B.3). Moreover, "when $\theta = \tilde{\theta}$, the sampling and scoring steps can be combined, and ... the detection task can be completed by only calling the model once" (Section 3.2).
- **Discriminative effect.** After sampling normalization, the distributions become much cleaner: LLM texts cluster around a mean score of ≈ 4, human texts around ≈ 0, with only minimal overlap (Appendix E, Figure 12, versus the flatter Lastde distributions in Figure 11). Appendix G additionally notes the Lastde++ decision boundary is consistently near **2.0** across source models.

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3. Experimental Setup

3.1 White-box vs. black-box scenarios (Appendix A)

- The task definition follows Fast-DetectGPT: given a text t, determine whether it is LLM-generated, *without any machine training* — the detector may use any available proxy model (Appendix A).
- **White-box:** the probabilities $p_\phi(t_i \mid t_{<i})$ of the *source model* M_ϕ that generated the text are accessible; the detector scores t with the source model itself. Suited to precise tasks like text/model attribution.

- **Black-box:** the source model is unknown/inaccessible; a fixed **open-source proxy model** (GPT-J by default) performs inference, sampling, and scoring on behalf of the detector, exploiting the fact that LLMs share similar architectures and learn common patterns of human text. Black-box is "more common in real-world scenarios" (Appendix A).

3.2 Datasets (Section 4.1, Appendix B.1)

- **Main English datasets (4):** XSum (BBC news documents), SQuAD (Wikipedia-based QA context), WritingPrompts (story generation), and Reddit ELI5 (QA limited to biology, physics, chemistry, economics, law, technique). Main white-box results use **XSum, SQuAD, WritingPrompts**; main black-box results use **XSum, WritingPrompts, Reddit** (Appendix B.1).
- **Cross-lingual datasets:** WMT16 (English–German translation) and a newly collected **Chinese Zhihu** dataset on food, history, and economics topics (Section 4.1).
- Each dataset has **150 human-written examples** (filtered to ≥150 words); the first **30 tokens** of each human text serve as the prompt for the source model's continuation, so each dataset ends up with 150 human (negative) and 150 LLM-generated (positive) samples of roughly equal length sharing a common prefix (Appendix B.1). PubMedQA was excluded for fairness because its samples are much shorter. System/user prompts for the closed-source APIs are documented in Table 4 (Appendix B.1).

3.3 Source models and proxy model (Section 4.1, Appendix B.2, Table 5)

- **18 source models in total:** 15 open-source models from 1.3B to 13B parameters — mGPT (1.3B), GPT-2 XL (1.5B), OPT-2.7B, GPT-Neo-2.7B, Phi-2 (2.7B), GPT-J (6B), Yi-1.5-6B, Falcon-7B, Gemma-7B, Qwen1.5-7B, BLOOM-7.1B, Llama-3-8B, OPT-13B, Llama-13B, Llama-2-13B — plus **3 closed-source models:** GPT-4-Turbo, GPT-4o, Claude-3-haiku.
- **Main results** use the 12 open-source models (excluding mGPT, Qwen1.5, Yi-1.5, which serve cross-lingual experiments). In the black-box main table, GPT-J is excluded as a *source* (it is the proxy) and **GPT-4-Turbo** fills that slot.
- **Proxy models:** GPT-J for English/German, Yi-1.5-6 for Chinese (Section 4.1, Figure 7). Fast-DetectGPT's original sampling (GPT-J) / scoring (Neo-2.7) split was unified to a single GPT-J for fairness (Appendix B.2).
- Hardware: two RTX 3090 (2×24 GB); models >20B (e.g., NeoX-20B) were not run locally (Appendix B.2).

3.4 Baselines and metric (Section 4.1, Appendix B.3)

- **8 training-free baselines in two groups:** *Sample-based* (single-score statistics): Log-Likelihood, LogRank, Entropy, DetectLRR; *Distribution-based* (contrast samples): DetectGPT, DetectNPR, DNA-GPT, Fast-DetectGPT. Lastde is benchmarked against the first group and Lastde++ against the second.
- Metric: **AUROC**, following the convention of prior work. Additional baselines (Binoculars, GPTZero) and third-party datasets (ReMoDetect, Fast-DetectGPT release) appear in Appendix D.

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4. Key Findings

4.1 Cross-model performance (Tables 1–2, Section 4.2)

- **White-box (12 source models, avg. over XSum/SQuAD/WritingPrompts):** Lastde averages **95.89%** AUROC vs. 90.80% for the best sample-based baseline DetectLRR (+5.09); Lastde++ averages **98.85%** vs. 97.82% for

Fast-DetectGPT (+1.03) — while needing only 100 samples instead of 10,000. Lastde leads by ≈8% on the LLaMA series and Gemma (Appendix C.1).

- **Black-box (11 open-source + GPT-4-Turbo, avg. over XSum/WritingPrompts/Reddit):** Lastde averages **86.38%** vs. 80.45% for DetectLRR (+6.38); Lastde++ averages **91.47%** vs. 85.68% for Fast-DetectGPT (+5.80). Notably, the single-score Lastde "outperformed all existing single-score methods, even matching the detection performance of Fast-DetectGPT" in the black-box setting (Section 1).
- All methods degrade from white-box to black-box, but Lastde/Lastde++ keep a consistent lead; for Llama3-8 the paper notes likely saturation, where fast-sampling underperforms the base statistics (Appendix C.2).
- **Closed-source generalization:** on GPT-4-Turbo (black-box) Lastde reaches 81.33 and Lastde++ 88.21 (Table 2); on GPT-4o and Claude-3-haiku, Lastde(Std) is the best sample-based method and appropriate aggregation functions (e.g., ExpStd) further help Lastde++ (Appendix D, Table 8).

4.2 Cross-lingual performance (Figure 7, Section 4.3)

- On **WMT16-German**, Lastde++ beats Fast-DetectGPT by an average margin of **+4.56%** AUROC across both scenarios; on **WMT16-English** by **+5.24%**.
- On the **Zhihu Chinese** datasets (Economy, Food, History), both methods perform well, with Lastde++ holding a slight advantage on all three topics.
- Conclusion in the paper: Lastde++ "is better suited for cross-lingual detection scenarios compared to Fast-DetectGPT."

4.3 Robustness against paraphrasing attacks (Table 3, Section 4.3, Appendix F.2)

- Attack setup: texts generated by source models are paraphrased with **T5-Paraphraser** (T5_Paraphrase_Paws), following Bao et al. (2024).
- Both Fast-DetectGPT and Lastde++ degrade under paraphrasing, **but Lastde++ consistently shows smaller drops**. Averaged across three source models:
  - White-box: XSum −8.48 vs. −10.60; WritingPrompts −1.73 vs. −2.70; Reddit −2.10 vs. −3.48 (Lastde++ vs. Fast-DetectGPT).
  - Black-box: XSum −9.14 vs. −9.45; WritingPrompts −4.20 vs. −5.44; Reddit −3.36 vs. −3.00.
- On paraphrased Reddit (GPT-4-Turbo), Lastde++ outright *surpasses* Fast-DetectGPT, "indicating superior robustness" (Section 4.3).

4.4 Additional robustness and analysis highlights

- **Sample efficiency (Figure 4):** Lastde++ with only **10** contrast samples already surpasses Fast-DetectGPT with **100** samples.
- **Response length (Figure 5):** detection improves with length; beyond 120 words Lastde/Lastde++ clearly outperform comparables. Short texts hurt because they restrict the number of usable scales τ.
- **Decoding strategies (Figure 6, Table 15):** under top-p=0.96, top-k=40, and temperature=0.8, Lastde++ stays above 94% AUROC and beats Fast-DetectGPT under every strategy — e.g., with top-k=40, average advantages of +8.30/+6.49/+3.70 across the three datasets.
- **Proxy-model sensitivity (Figure 8):** the proxy choice strongly affects black-box detection; Lastde++ wins in all five (source, proxy) pairings tested, including the hard GPT-4-Turbo + Llama3-8 pairing where all methods perform poorly.
- **Third-party datasets and strong baselines (Appendix D, Tables 9–10):** on ReMoDetect/Fast-DetectGPT data (NeoX-20B, Llama3-70B, GPT-3.5-Turbo, GPT-4, GPT-4-Turbo, Claude-3; domains

XSum/SQuAD/WritingPrompts/PubMedQA) and against Binoculars and GPTZero, Lastde++ averages **94.2%** AUROC (vs. 93.7% Fast-DetectGPT) even with non-optimal unified hyperparameters; combined with the TOCSIN plugin, **95.3%** vs. 94.3%.

- **Human-style mimicking (Appendix F.4, Table 16):** on GPT-4-Turbo/GPT-4o texts prompted to "mimic the human writer," Lastde++ matches Fast-DetectGPT on AUROC (96.49) and wins on TPR@5%FPR (85.36 vs. 82.67) and Accuracy@5%FPR (90.17 vs. 88.84).
- **LLM–LLM mixture text (Appendix F.5, Tables 17–19):** on truncation-concatenated mixtures of 2 and 4 LLMs, Lastde/Lastde++ reach SOTA on all three metrics, with TPR@5%FPR exceeding Fast-DetectGPT by at least 20%.
- **Threshold in practice (Appendix G):** with logistic-regression thresholds, Lastde++ averages 91.40% AUROC and 52.33% TPR@1%FPR (vs. 86.00% / 32.62% for Fast-DetectGPT). The paper acknowledges that label-based threshold fitting "appears to undermine the 'training-free' aspect" and proposes a label-free pseudo-label scheme around the ≈2.0 score boundary.

4.5 Limitations acknowledged by the authors

The Ethics Statement and Appendix G concede: detection "still produce[s] imperfect detection results when faced with sophisticated challenges such as paraphrasing attacks and the arbitrary replacement of proxy models," and detection scores should support, not replace, human judgment.

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5. Items Not Reported in the Paper

- **No comparison with training-based detectors** on the six datasets — all 8 main baselines are training-free (Section 4.1).
- **No statistical significance testing** of AUROC differences is reported.
- **No absolute wall-clock/latency measurements** for Lastde/Lastde++; efficiency claims are relative (100 vs. 10,000 samples; the 340× speedup of Fast-DetectGPT over DetectGPT is cited from Bao et al., 2024, not measured here).
- **The mechanics of fast-sampling are not re-derived** in the paper; they are cited from Fast-DetectGPT (Section 3.2).
- **Hyperparameters for closed-source experiments** (Table 8: $\varepsilon=1$ for Lastde; unified $\varepsilon=10$, $\tau'=5$ in Appendix D) deviate from the main defaults without a detailed justification beyond the brief discussion of aggregation functions.

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6. Overall Assessment

The paper's core contribution is conceptual and empirical: it reframes the TPS as a **time series** and shows that **multi-scale local-dynamics features (MDE), when fused with the global likelihood, materially improve training-free detection**, especially in the harder black-box, cross-model, cross-lingual, and paraphrasing settings — at a fraction of Fast-DetectGPT's sampling cost. The methodology is fully self-contained (Eqs. 1–11, Figure 2), the evaluation is broad (18 source models, 6+ datasets, 3 languages, 8–10 baselines, third-party data, ablation, and attack studies), and the limitations are candidly stated.